



Electrode Placement Systems and Montages

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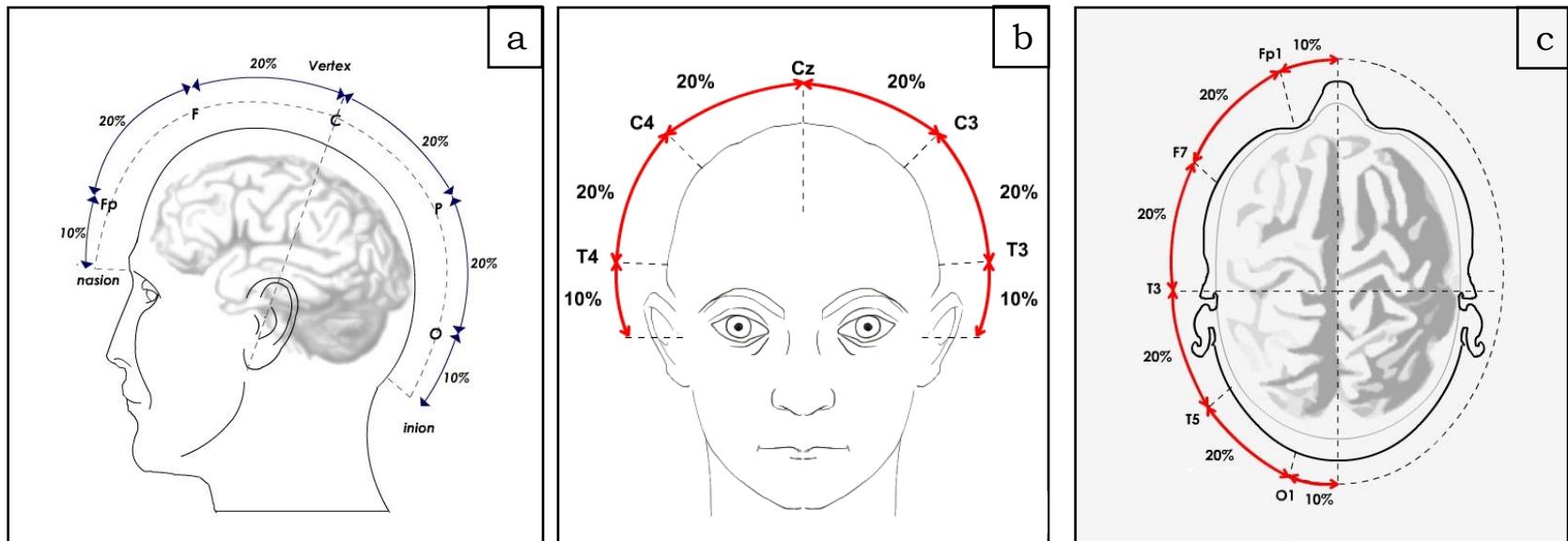
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Traditional International 10-20 System

When Hans Berger recorded the first human EEG (1924) he only had two electrodes available, and he positioned them in the anterior and posterior regions of the head. Berger kept using this method for many years, considering it an efficient system to measure global cortical activity.

Later on, other researchers highlighted how, in reality, EEG activity varied significantly depending on the area of the scalp from which it was recorded. Observation of different regional kinds of cerebral rhythms then encouraged the use of multiple electrodes and of more recording channels, but standardization of the recording methods soon became necessary, so that the resulting data could be comparable with one another. A committee of International Federation of Societies for EEG and Clinical Neurophysiology (IFSECN), led by H. Jasper, started working on a specific electrode positioning system, to be used in all laboratories.

The first standardized system was presented at the 2nd International Congress of IFSECN in Paris in 1949 and published by Jasper in 1958; it is still universally used and known as the International 10-20 System (SI 10-20).

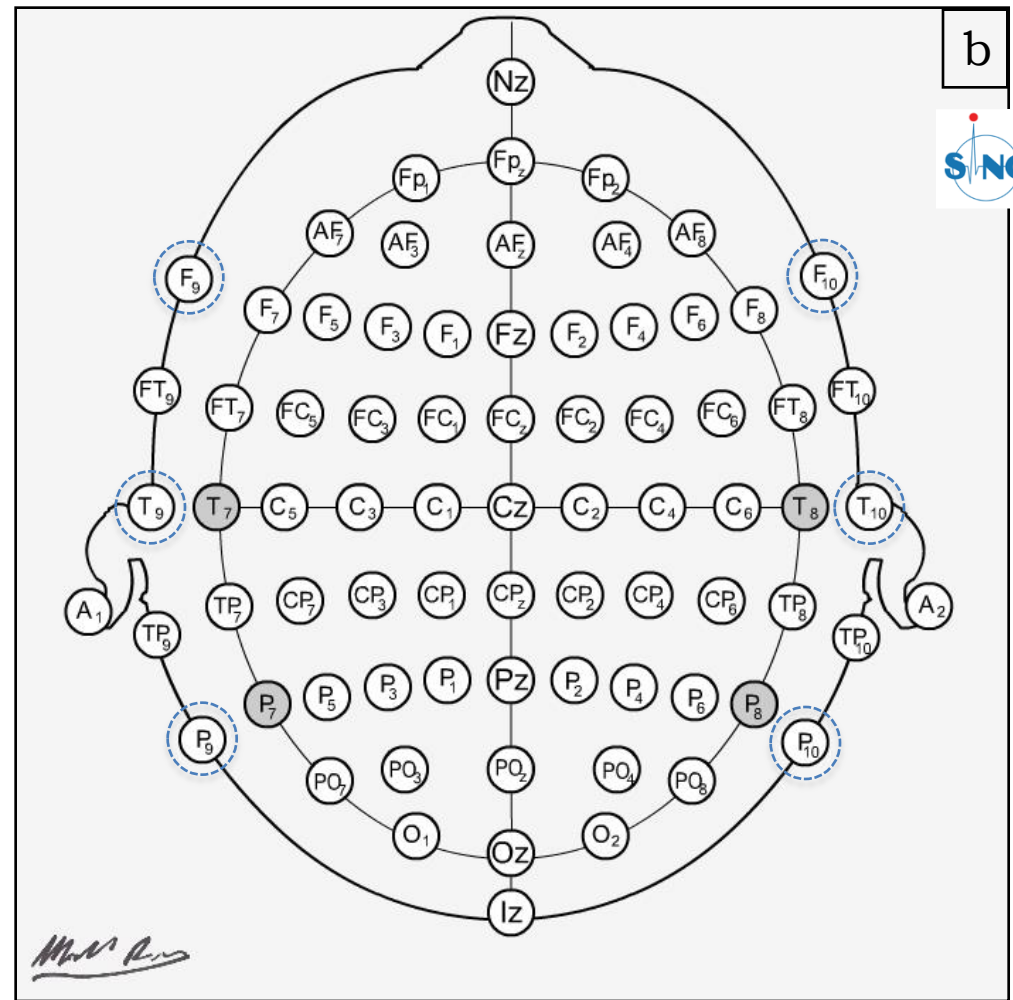
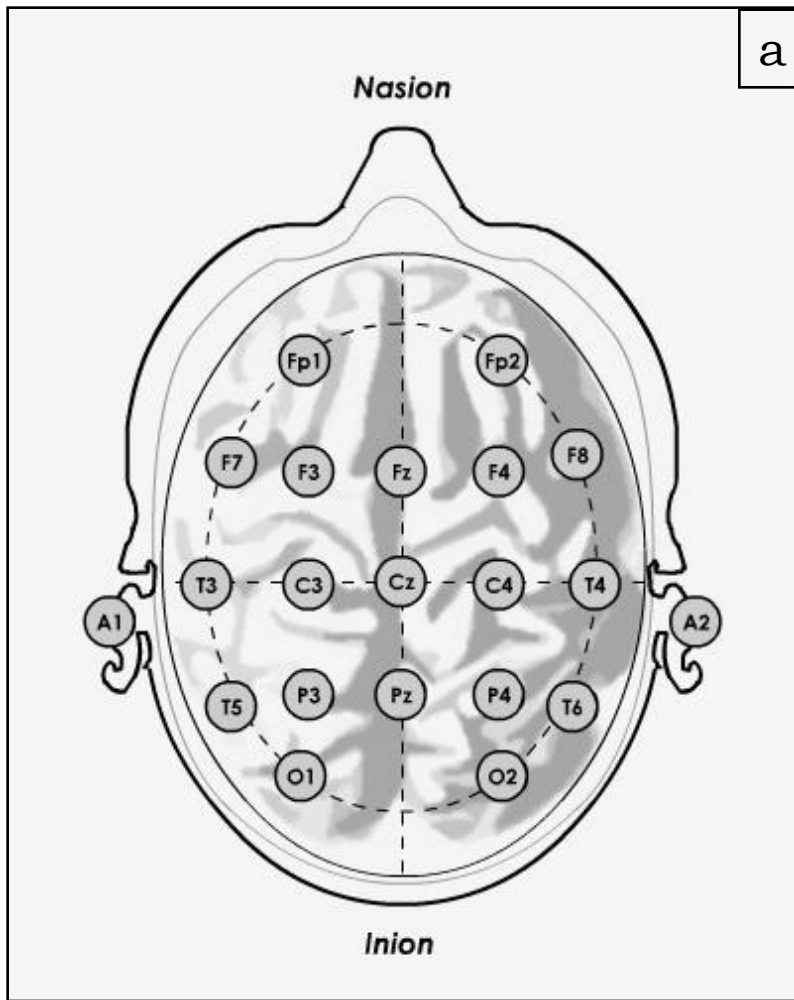


Traditional 10-20 system.

- a*: antero-posterior mesial line, connecting nasion and inion;
- b*: latero-lateral coronal line, connecting the two pre-auricular points, through the vertex;
- c*: sagittal lateral longitudinal line, connecting nasion and inion.

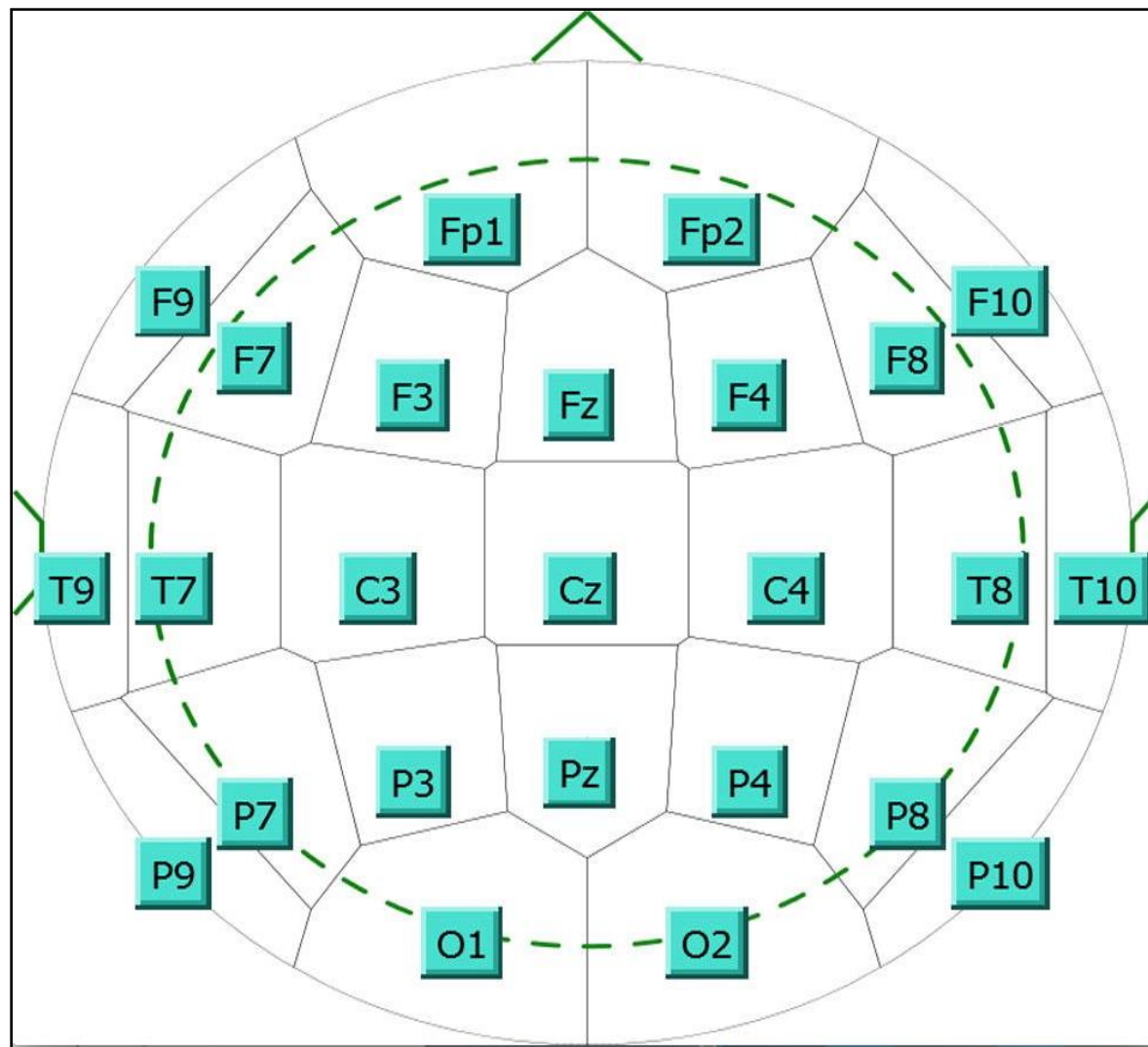
Modification of 10-20 system (10-10 system)

- The development of digital EEG and the introduction of high density EEG and source localization methods made it necessary to increase the electrode arrays.
- Therefore a modification of 10-20 nomenclature with the definition of 10-10 combinatorial nomenclature has been proposed and accepted by the American Clinical Neurophysiology Society (ACNS) and by the International Federation of Clinical Neurophysiology (IFCN).
- The modified combinatorial nomenclature is an extension of the 10-20 system and it entails the positioning on the scalp of > 70 electrodes, placed along 11 sagittal chains and 9 coronal chains.
- The modified 10-10 terminology replaces the inconsistent T4/T3 and T6/T5 terms with the consistent T8/T7 and P8/P7 .

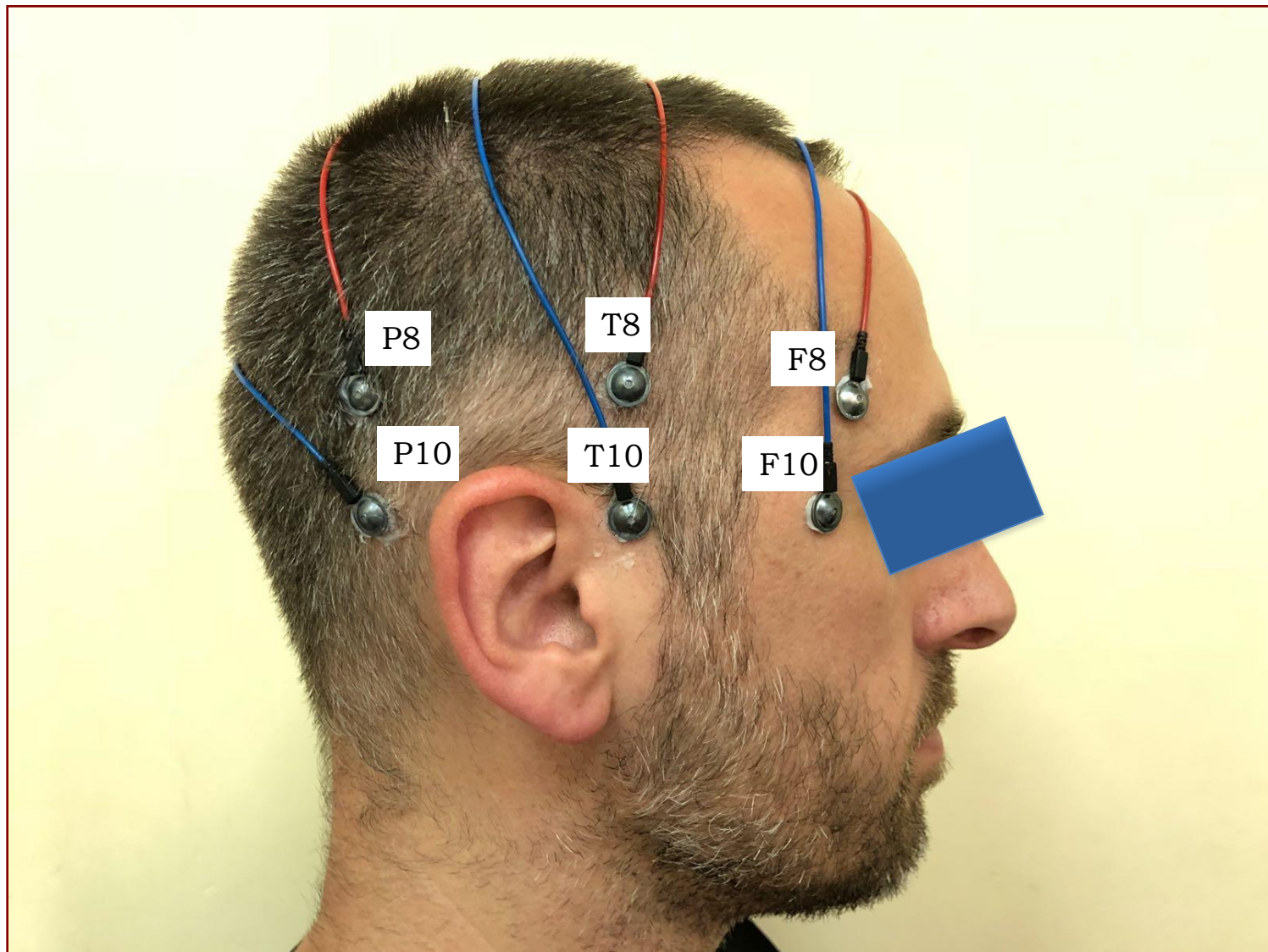


a: standard positions of 19 scalp electrodes and two ear reference electrodes, according to traditional 10-20 system; **b**: modified 10-10 system, with 73 electrode positions on the scalp and two ear reference electrodes (note that the electrodes T8, P8, T7, and P7 replace the electrodes formerly named T4, T6, T3, and T5; the electrodes of the inferior temporal chain F10, T10, P10, F9, T9, and P9 (dotted circles) are actually recommended as new standard montage with 25 electrodes

- During the routine recordings with the traditional 10-20 system the placement of 19 standard scalp electrodes does not always detect the activity originating or propagating from mesial temporal structures.
- For this reason the IFCN recommends a new standard array for clinical practice that includes the six electrodes of inferior temporal chain (F10/F9, 10% inferior to F8/F7; T10/T9, 10% inferior to T8/T7; and P10/P9, 10% inferior to P8/P7): it results in a total of 25 electrodes placed on the scalp (Fig. 3, Fig. 4).⁷
- Recently it has been demonstrated that adding six electrodes in the inferior temporal chain to the traditional 10-20 system improves the identification of EEG abnormalities originating from the basal part of the temporal lobes.



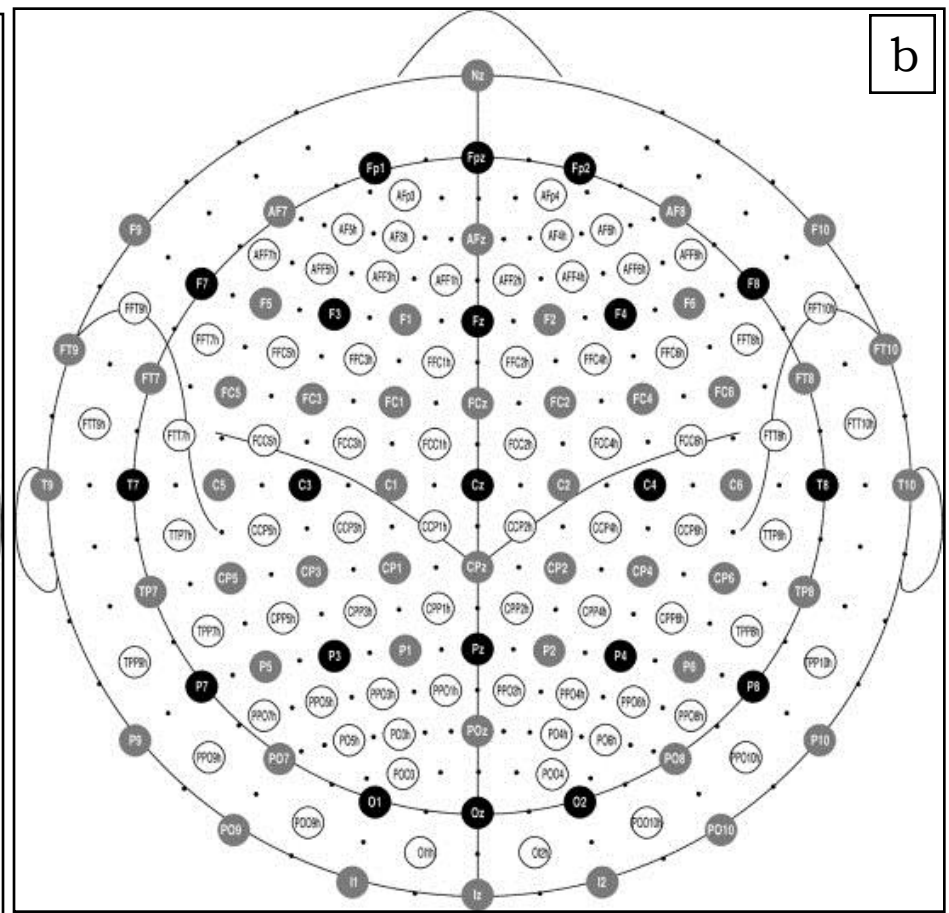
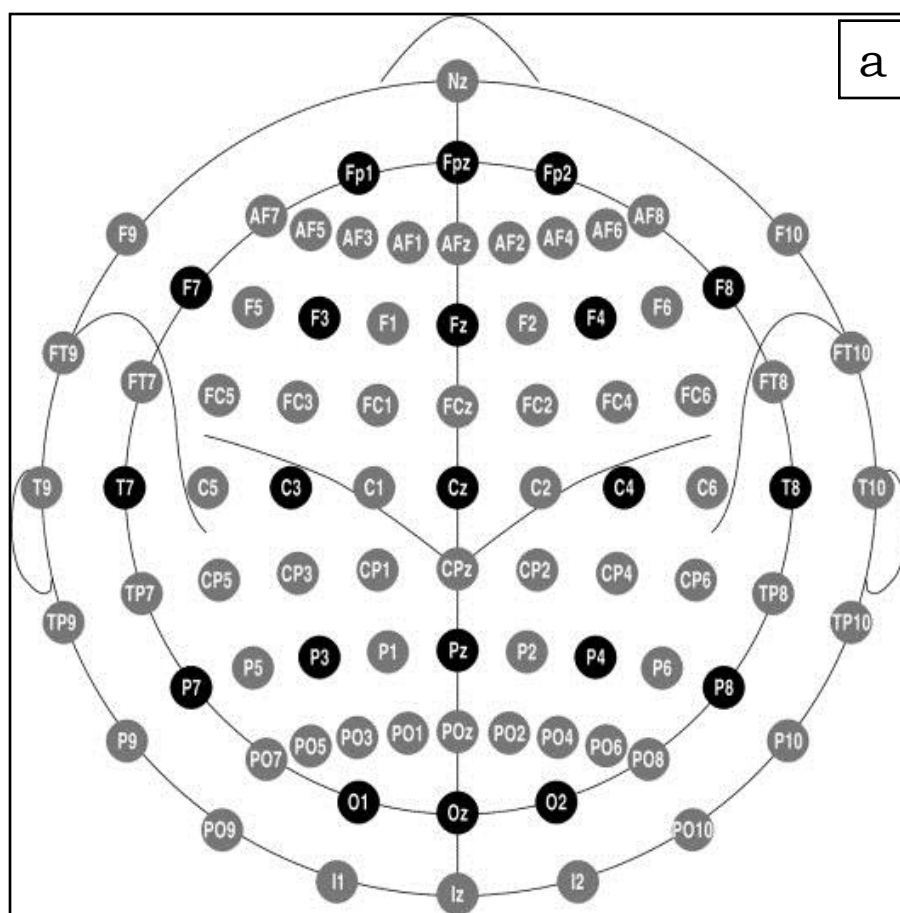
New standard montage with additional coverage of the inferior and anterior brain regions, according to the recent recommendations of International Federation of Clinical Neurophysiology (from Seeck M, Koessler L, Bast T, et al. The standardized EEG electrode array of the IFCN. Clin Neurophysiol. 2017; 2070-2077)



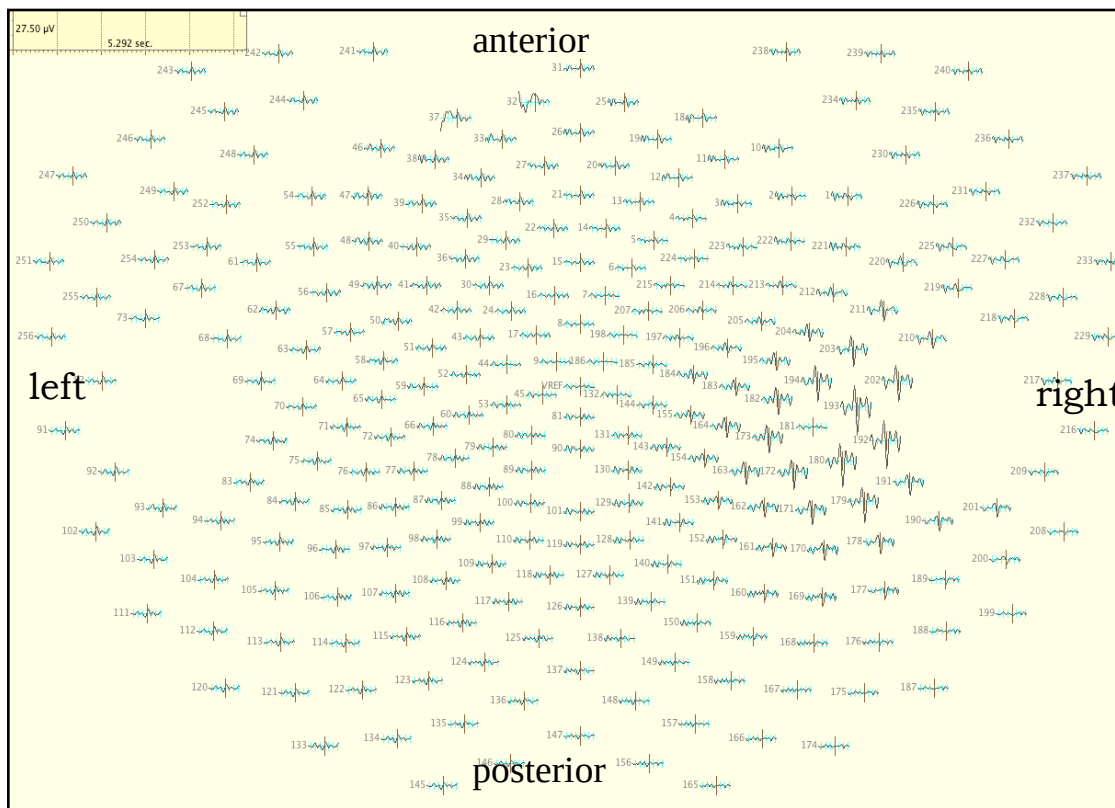
Electrodes of the right inferior temporal chain according to new recommendations of IFCN (T8 = ex T4; P8 = ex T6).

Proposed 10-5 system for High-resolution EEG

- A further extension of 10-10 system (named 10-5 system) was proposed in 2001 but it has not yet been accepted by ACNS and IFCN. The 10-5 proposed extension defines the position and nomenclature of 345 locations on the head, and it can accommodate a homogeneous distribution of a subset containing, for example, 128 or 256 electrodes.
- The nomenclature of this system utilizes the combination of two letters to indicate the contours lying halfway between the original 10-20 system contours (the electrodes between the F and C contour were labelled FC, and so on for others).
- In this way, the locations for the coronal contours from anterior to posterior were named: AF, AFF, F, FFC, FC, FCC, C, CCP, CP, P, PPO, PO. Electrodes for High Density EEG are applied by using expandable nets or caps with embedded electrodes and their localization is determined by digitization in three-dimensional space.

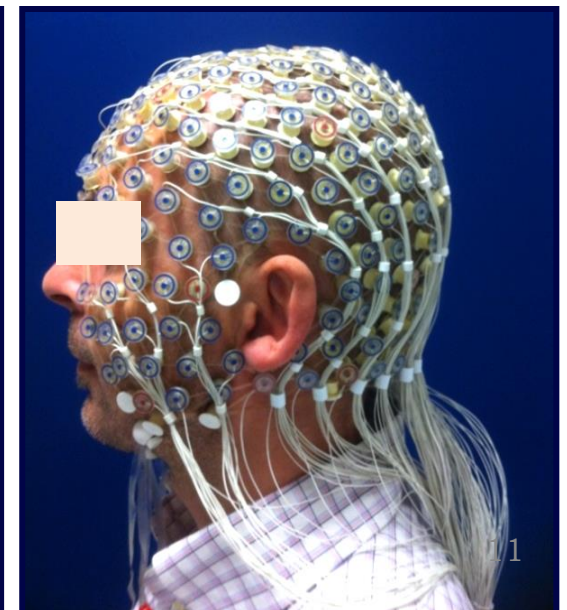
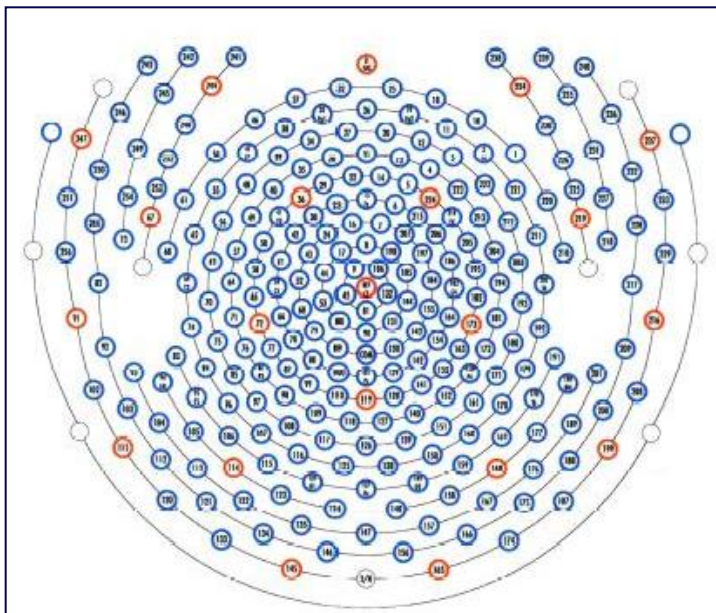


The 10-10 and 10-5 extension of traditional 10-20 system. In **a** black circles indicate positions of the original 10-20 system, and gray circles indicate additional positions introduced in the 10-10 extension. In **b** electrode positions in the proposed 10-5 system: additional positions to the 10-10 system are indicated with dots; a selection of additional positions useful for a 128 channel EEG system are indicated with open circles (from Oostenveld R and Praamstra P. The five percent electrode system for high-resolution EEG and ERP measurements. Clin Neurophysiol. 2001; 112: 713-719)



An example of 256 channels high density EEG, with projected locations of electrodes on the scalp

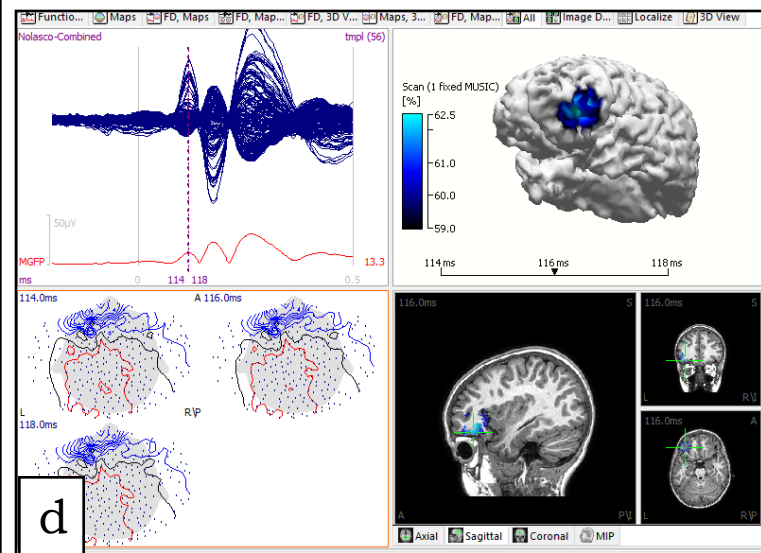
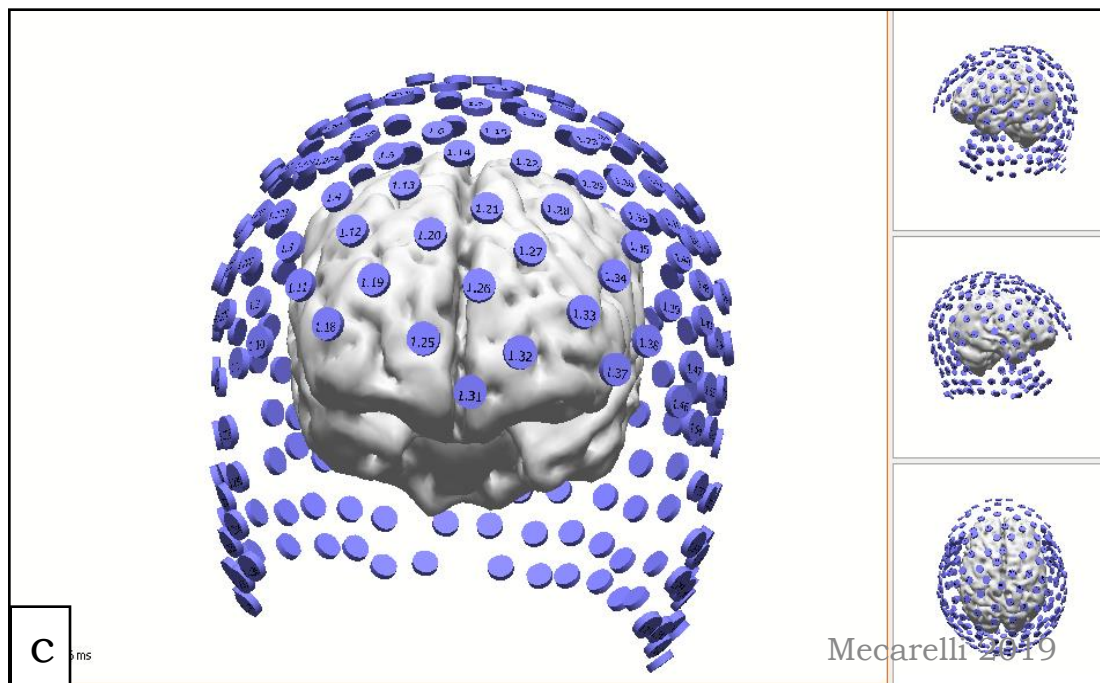
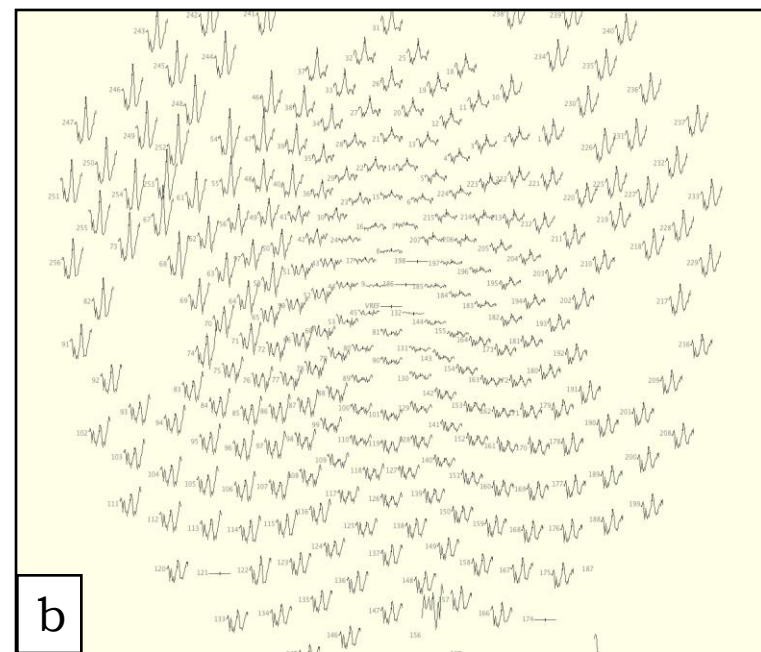
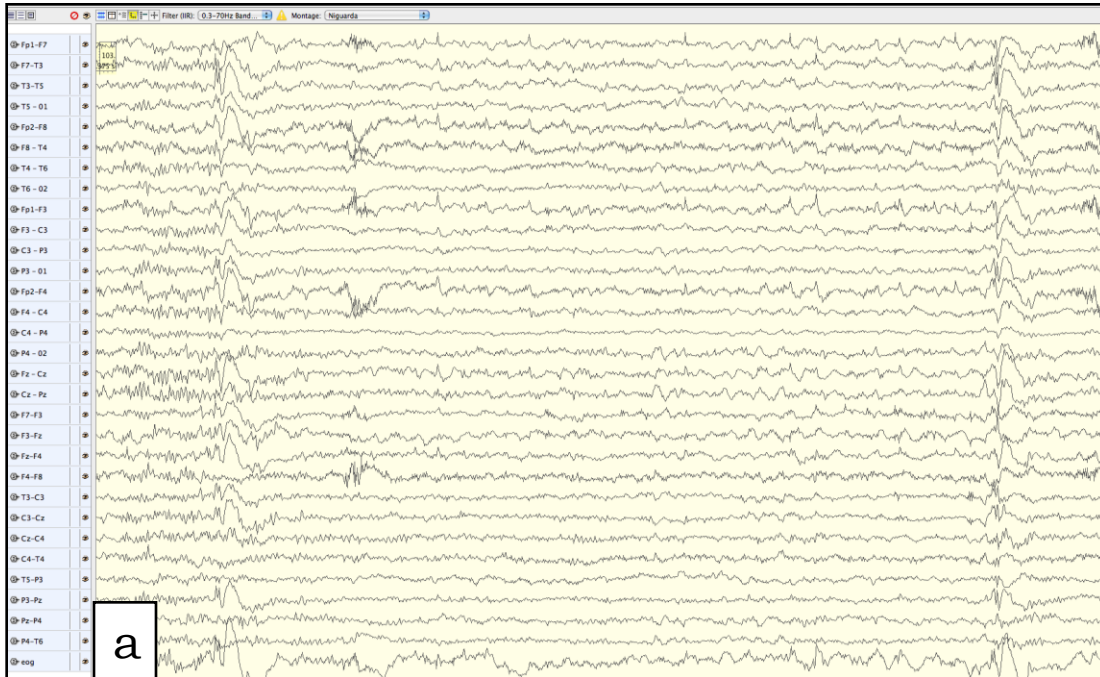
(courtesy from: Paolo Manganotti, Clinical Neurology Unit, University of Trieste – Italy)



Legend of the following Image

Comparison of standard and high density EEG in a patient with epileptic left temporal focus; a: standard EEG recording with placement of electrodes according to 10-20 system; b: recording of a single spike by 256 electrodes placed on the scalp; c: the 256 placed electrodes projected onto a 3D image of the patient's brain, obtained by MRI; d: source analysis of epileptic focus (56 spikes average)

(courtesy from: Annalisa Rubino, Lino Nobili, Epilepsy Surgery Centre – Niguarda Hospital, Milan, and Child Neuropsychiatry, Dept of Neurosciences, University of Genoa)



Final recommendations

Although in clinical practice the electrode placement method should be individualized on the basis of the clinical needs of the individual patient, the following recommendations ought to be kept in mind, according to the Guidelines of ACNS and IFCN:

- the 10-20 traditional system (21 electrodes) may be adequate for most patients, also for ambulatory or video-EEG longterm monitoring;
- in the suspicion of epilepsy or in epileptic patients without a clear visualization of the epileptic focus, supplementing the 10-20 EEG array with six electrodes for the inferior temporal chain (25 electrodes in total) is recommended;
- in children, except for special cases, the same number of electrodes as in adults is usually recommended;
- the 10-10 system should be used in epileptic patients undergoing pre-surgical evaluation, and for source localization purpose;
- for the transition from the traditional system to the new larger arrays, the modification of EEG machine headboxes and a gradual process of educating operators on the new terminology are necessary.

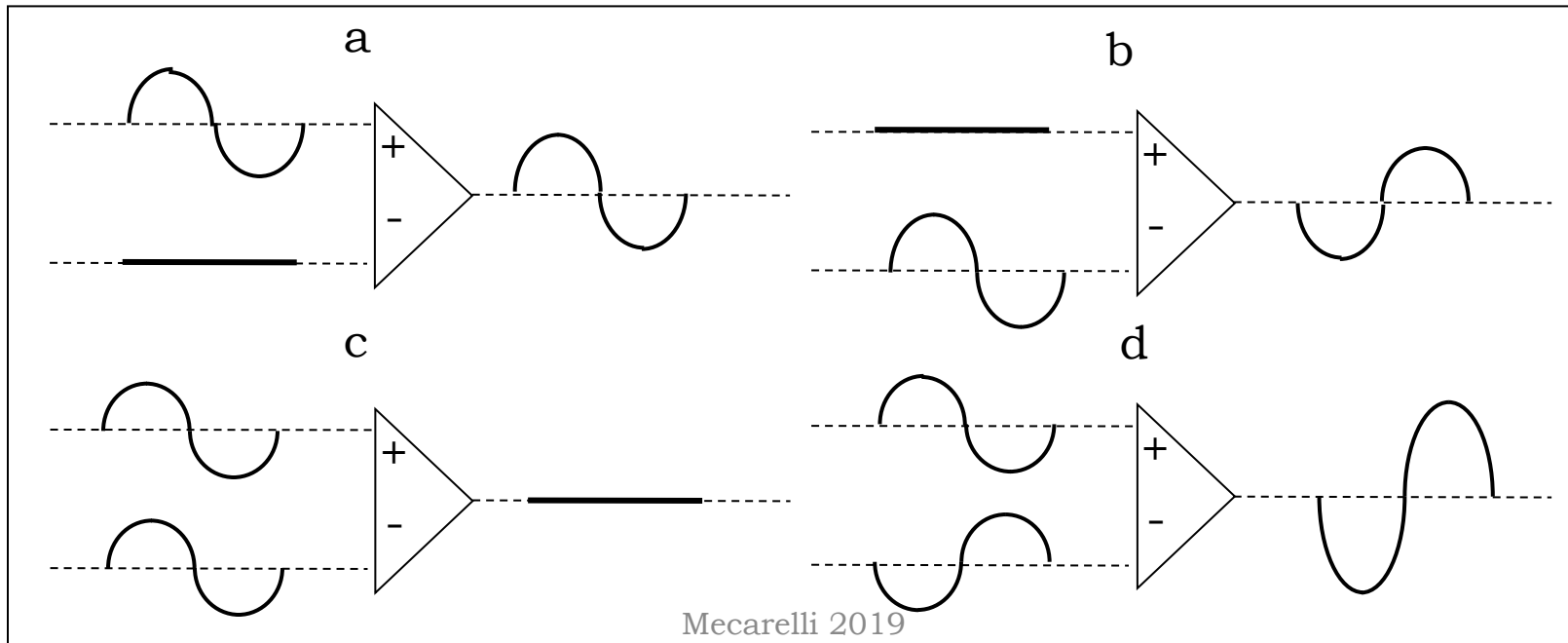
Electrode derivations and montages

- Brain electrical signals are displayed on the monitor depending on how the electrodes are connected to the amplifiers.
- Each amplifier has two electrode inputs (1 and 2), and then the *potential measurement* represents the *potential difference* between these two points. From the electrical point of view, one of these points is called “common” or “ground” and its potential is deemed to be *zero*: unfortunately, ground points generate potentials and do not represent a real zero.
- This can be overcome by connecting two amplifiers together at the ground point and comparing the potential difference between their active inputs.
- Then if two amplifier inputs (1 and 2) have the same polarity and voltage, the output will be zero (*in-phase rejection*), while when these potentials are different, the recorded output value will be proportional to the difference between the two input values.
- Using differential amplifiers, only the difference between two inputs is known, not the absolute value of the potentials of electrodes attached to either inputs 1 or 2.

Comparing a large number of electrode positions, the EEG allows, with a good approximation, the localization of an abnormal activity on the scalp.

In summary, on the monitor or recording paper, the signal is displayed with two fundamental characteristics: the voltage and the polarity. With regard to the polarity, by convention, if the relative voltage difference is negative, the signal deflects upward and, if the voltage difference is positive, it goes downward.

Therefore, if input 1 is more negative than input 2 the output signal will deflect upward, whereas if input 1 is more positive than input 2 the output signal will deflect downward. Finally, when inputs 1 and 2 have the same polarity and voltage, the output signal will be a flat line.



Reference Derivations

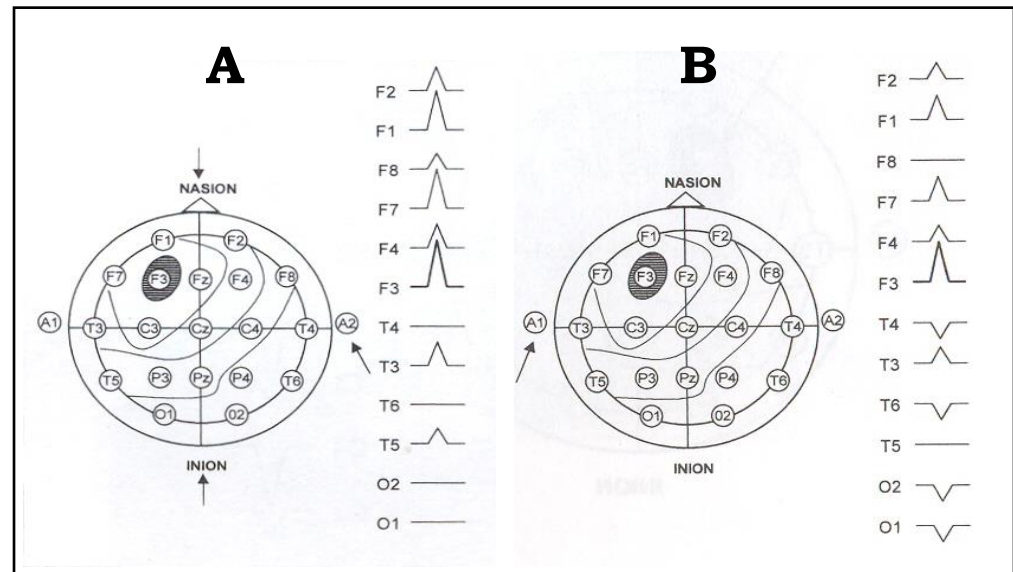


1. Common Reference

With this recording method, each electrode placed on the scalp is referred to as a common electrode placed at a point x, on the scalp or elsewhere.

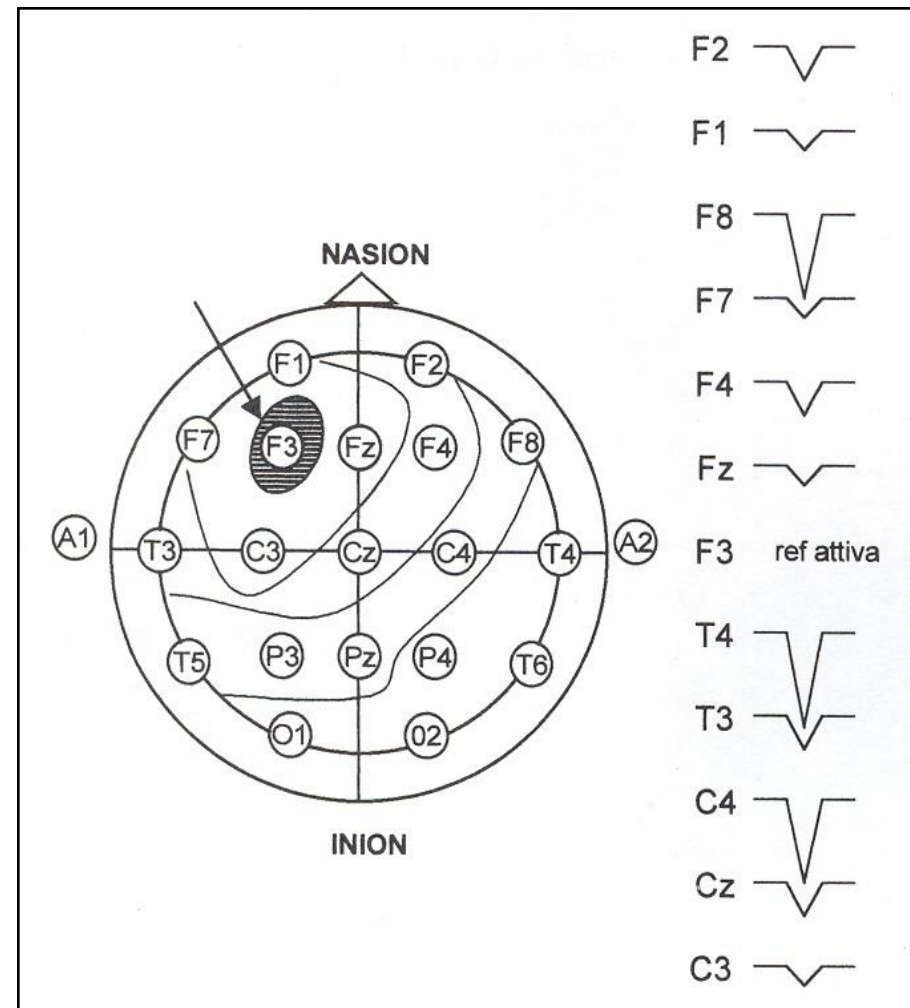
The common reference electrode should be as neutral as possible from the electrical point of view, which is a really rare occurrence.

The figure shows the fundamental principles of the so-called “inactive” common reference. A potential with a negative voltage peak at the F3 electrode generates a surrounding electric field with gradient marked by the circular lines. Using A2 as a reference, the signals in this virtual situation would be displayed like in A.

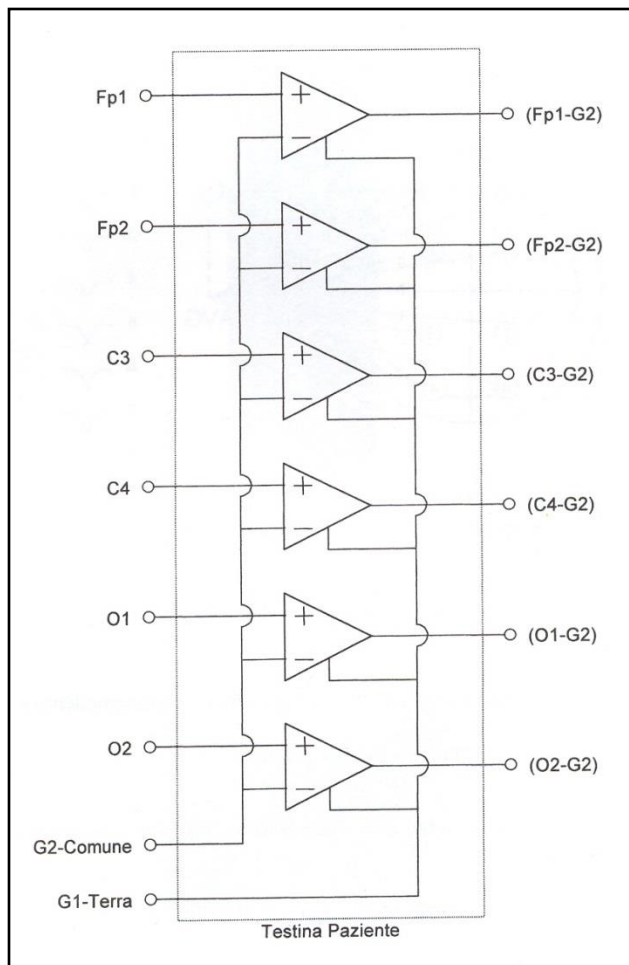


Choosing the A1 left ear electrode as a reference, which is located inside the F3 electrical field, and which has a low amplitude negative potential compared to the arbitrary zero (a common occurrence in normal practice), a very different pattern will be drawn like in B.

As a common reference, an active scalp electrode can also be chosen (**active common reference**). The figure shows a virtual situation in which the electrode placed in F3 is the one chosen as the common reference. F3 is over the maximum electrical field on the scalp and records the highest negative signal. Since in this case the reference is negative in comparison to other electrodes, all amplifiers will produce downward deflections. Moreover, since the electrodes surrounding the activity peak (F1, Fz, F7, C3) differ only slightly in potential from the reference, their corresponding channels will show smaller deflections than the ones farther away (the bigger the distance from the reference, the bigger the deflection).



Current digital EEG systems always refer the cerebral bioelectric signal to a common reference electrode, which typically has an input in patient head-box named G2 and which can be positioned on the scalp or elsewhere (usually it is placed medially on the scalp, anterior Fz or between Fz and Cz). The G2 common electrode is the common point referring to which all potentials of the single electrodes are measured



G2 - Common Reference recording. G2 electrode is connected to the inverting input (negative) of all amplifiers, while the electrode to be measured is connected to the non-inverting input. What the machine physically measures then is the potential difference between each electrode and G2 (Fp1- G2, etc).

2- Common Average Reference

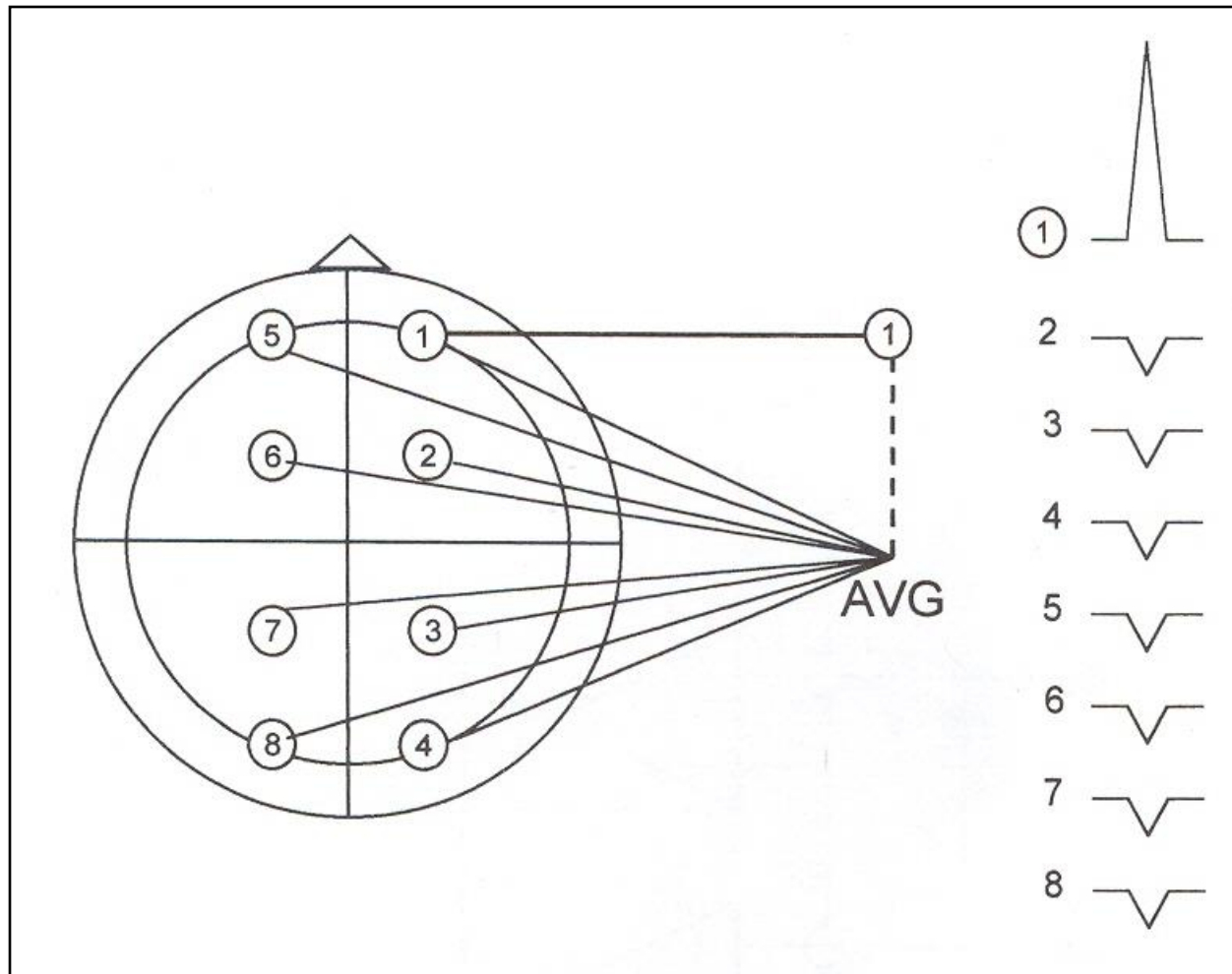
Many of the problems encountered with the use of Common Reference can be overcome by-Average Reference (AVG), a mathematical reference, introduced in Electroencephalography by Goldman and Offner in 1950.

In this case, the potentials of the single electrodes are referred to an instant average value obtained by adding together the potentials of all applied electrodes. The higher the number of electrodes is, the closer to zero the reference average potential will be.

One of the properties of the mathematical average of a series of numerical values that is the sum of the average differences which equals zero. Regarding the results on the EEG tracings, this means that we will always have positive or negative deflections, with respect to the zero value of the reference.

In analogical systems this derivation method is achieved by joining all electrodes to a high resistor placed in a common place inside the device.

Modern digital systems calculate the mathematical average of the potential differences of all the electrodes used, at the same time and moment by moment.

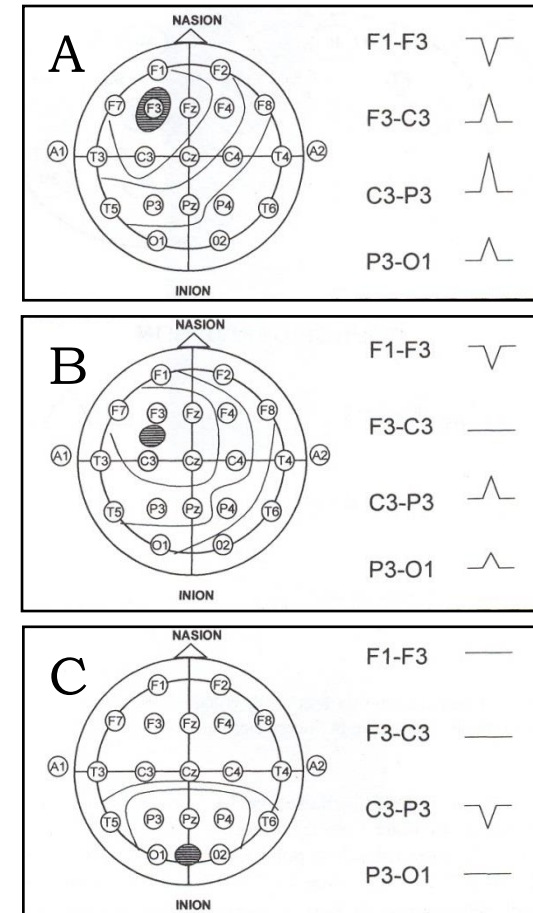


Average Reference (AVG) recording. In this case an ideal situation is presented, with only 8 electrodes applied to the scalp; electrode 1 has a more significant potential difference than the others (- 80 mV), while the other electrodes remain at arbitrary values of zero.

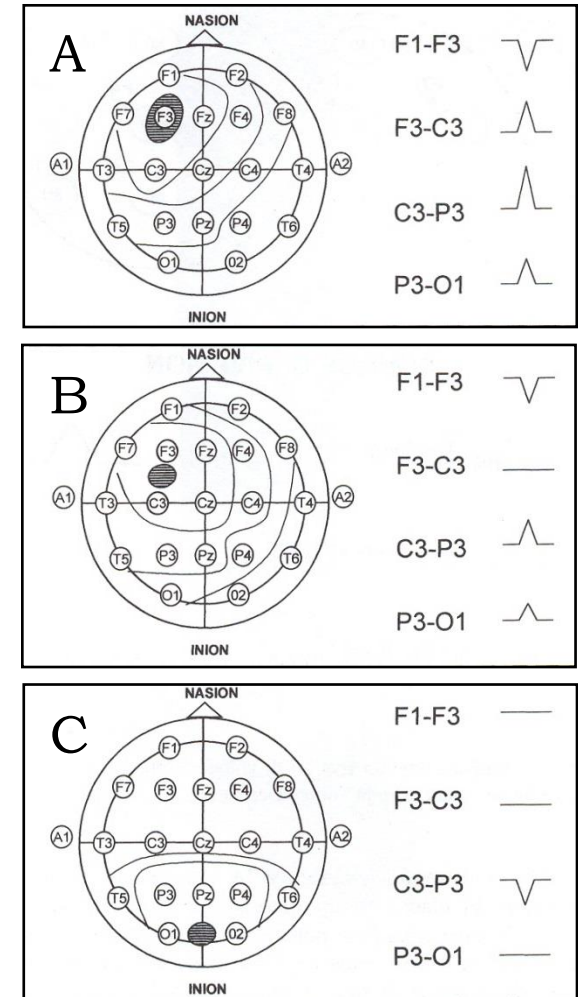
Bipolar derivations

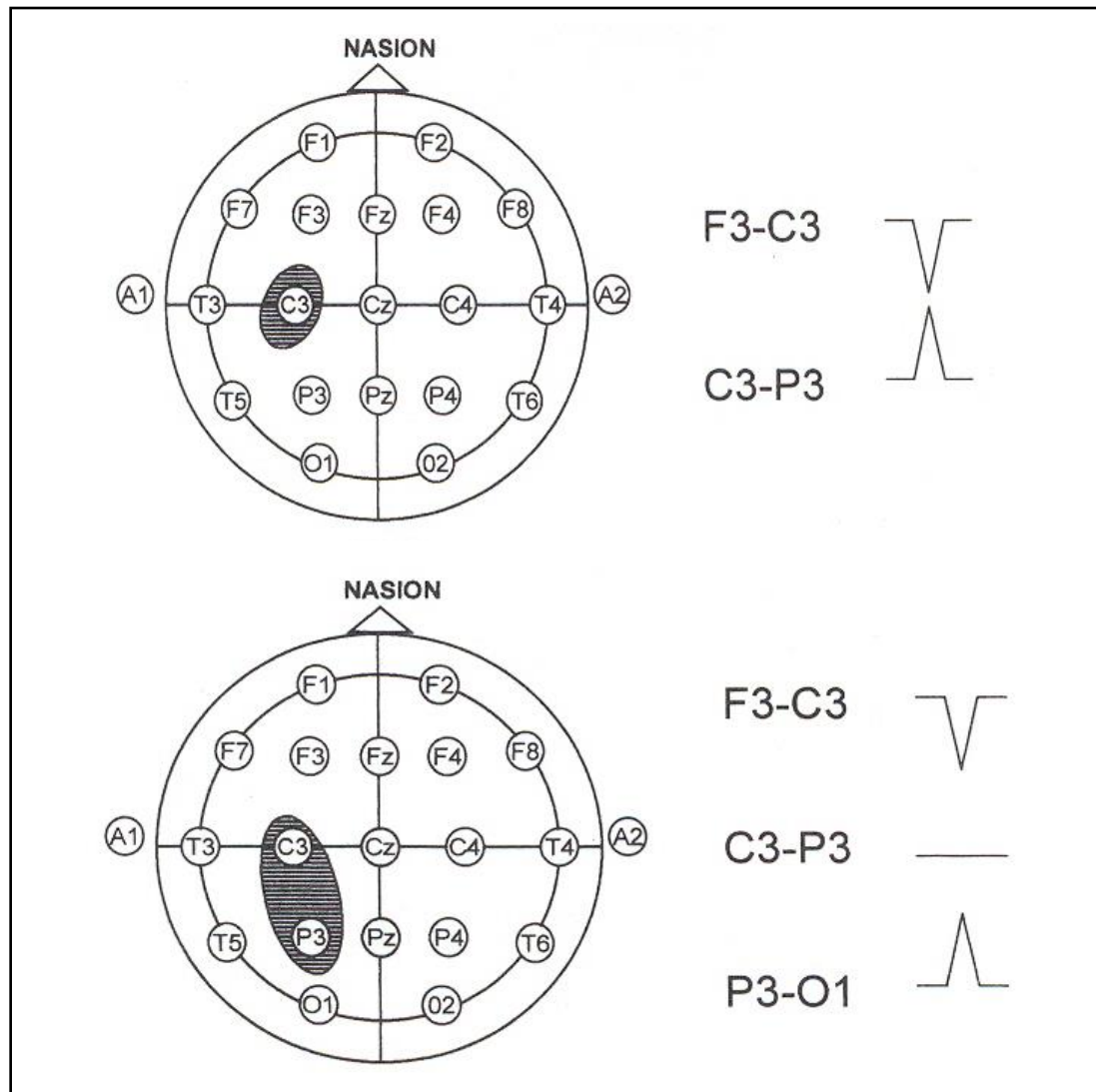
- In bipolar derivations, the potential difference is calculated between electrode pairs, placed along chains (longitudinal or transversal) in which an electrode is shared with two following channels.
- In this way, an event localized underneath a specific electrode will generate a deflection with the same voltage but opposite polarity in the two adjacent points in the chain of the electrode.

The figure in **A** shows how the previously described theoretical example of Common Reference (maximum negative potential at F3 electrode) appears using bipolar derivation. The phenomenon known as *phase-reversal*, typical of bipolar montages and caused by the fact that an electric event affects the electrode in the middle of the chain, occurs between channels 1 and 2. In this case F3 localizes the maximum field and the reverse phase is due to the fact that it is a common electrode to the first and second channel. This is an *instrumental phase reversal* because it is the instrument that causes the phase reversal and it must be distinguished from a *true phase reversal*, due to the fact that two different polarities are simultaneously present in adjacent cortical areas.



Where instead the maximum potential field is equally distant from the F3 and C3 electrodes (**B**), the second channel will not record any potential difference (zone of isopotentiality), while the phase reversal will be observed between the first and the third channel. Finally, if the maximum potential field takes place at the end of the chain (**C**) the only positive deflection will be detected in the channel connecting C3 with P3, possibly causing a wrong interpretation of the phenomenon.





Schematic representation of an instrumental phase reversal phenomenon between F3-C3 and C3-P3 channel. A maximum field potential underneath C3 will be observed in phase reversal on the two channels having C3 in common (*above*), while, if the event equally affects the two adjacent electrodes C3 and P3, the potential will be cancelled on the channel connecting these two electrodes (*below*).

Choice of derivation in clinical practice

Since EEG patterns are variable (focal or diffuse, transient or persistent) there is not one single ideal derivation to highlight all activities.

A first important factor to take into account in the choice of derivation is *interelectrode distance* and this is valid particularly for active common references and bipolar derivations. In bipolar derivations, distances between the paired electrodes are small and equal and this favours the detection of the fast EEG activities; in active common reference, distances are bigger and unequal and this facilitates the amplification of the signal, thus better showing the slower activity.

Focal epileptic activity is a phenomenon that provokes a change in the potential limited to a small part of the scalp. In order to accurately localize a strictly isolated focal potential with **bipolar derivations** it is necessary to demonstrate the simultaneous phase opposition at the point of intersection between two electrode chains placed perpendicular to each other: therefore it is necessary to use a suitable montage and, sometimes, to apply additional electrodes.

Common reference does not present such limitations and it has some advantages, especially for the detection of an epileptic activity with a not very good localized focus; it is important though to be sure that the chosen reference is not contaminated by the activity of the phenomenon itself (for example, in case of a focus with centrotemporal spikes, the ideal common reference is placed on the contralateral auricular lobe).



Average reference seems to be the preferable technique, even if the problem of contamination still exists for all channels (it is thus advisable to calculate the correct average reference, obtained by excluding the most active electrodes from the average calculation). When the focus is recorded by multiple electrodes, the phenomenon is better showed using common reference (preferably inactive), which provides a better definition of the potential distribution and of the shape of the wave.

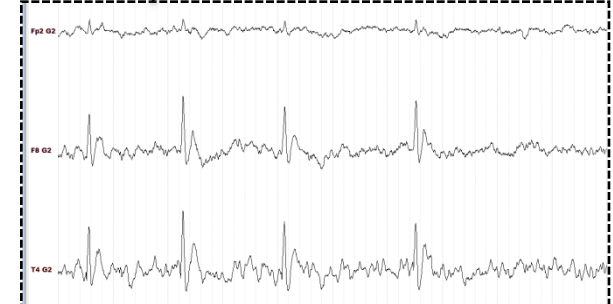
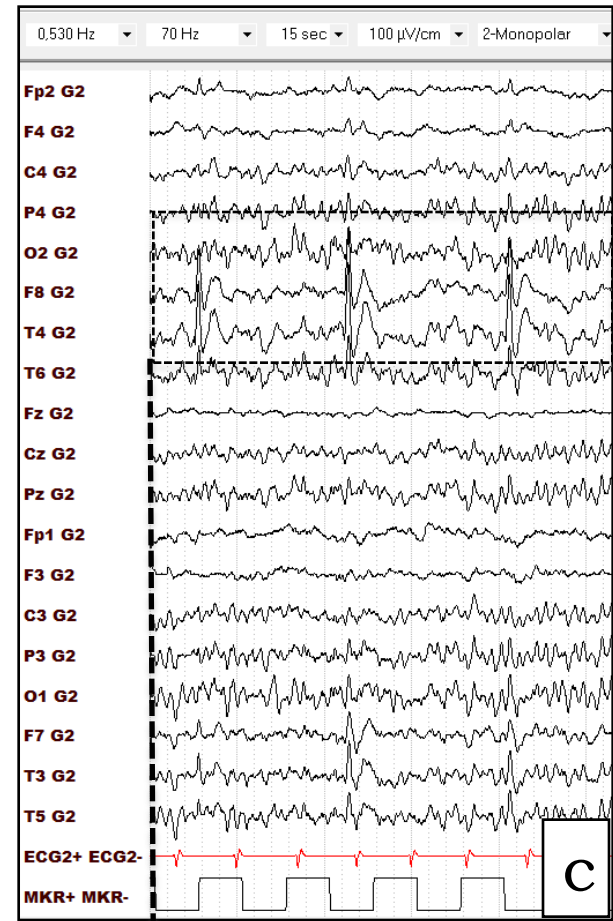
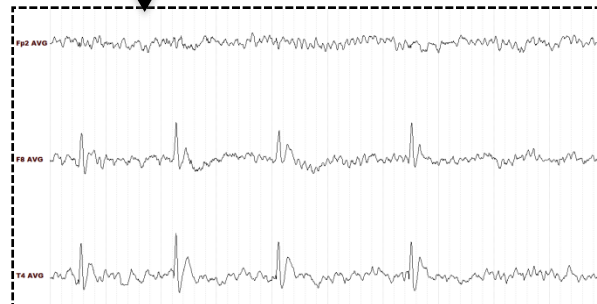
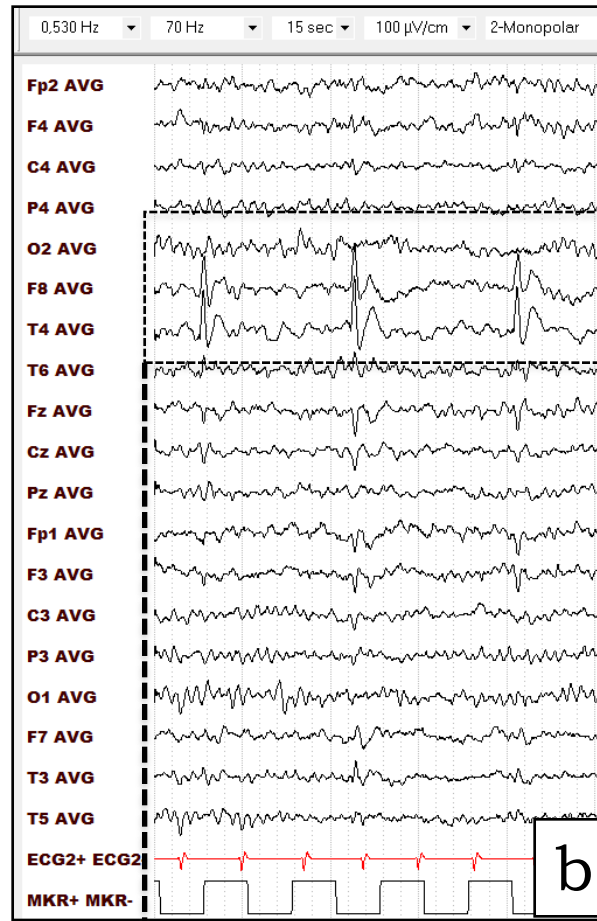
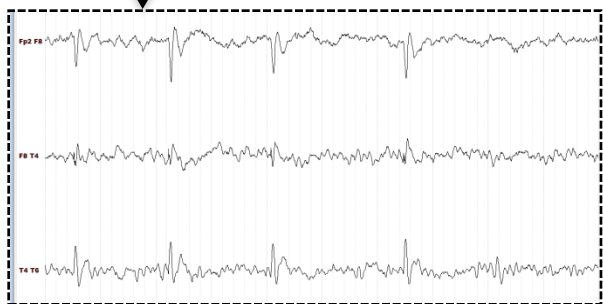
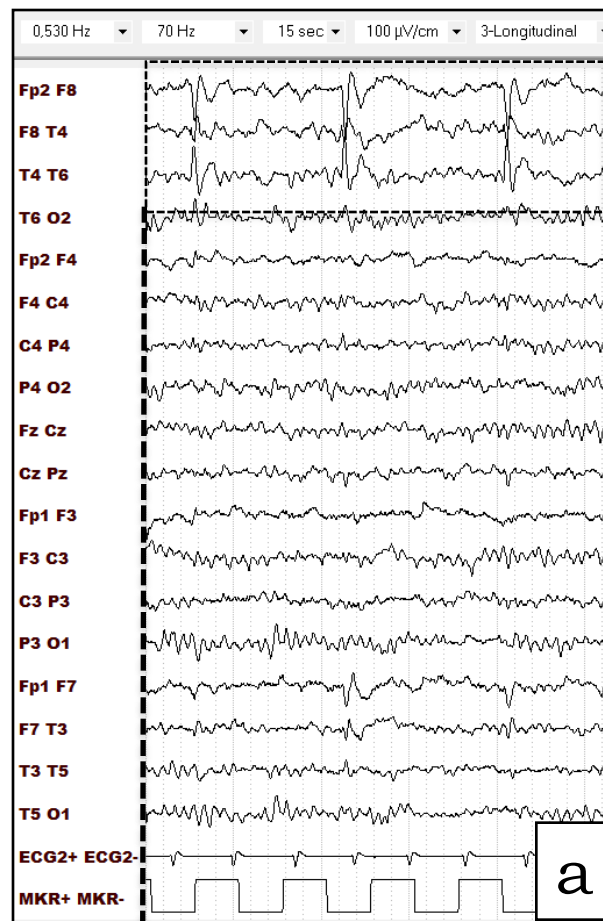
In order to correctly identify and localize any **focal activity**, more than one kind of derivation it should be used.

It is important to consider that:

- common reference allows to locate focal activity with higher voltage and with the same polarity;
- average reference shows a wider signal underneath one or more electrodes, but of opposite polarity than the majority of other electrodes;
- bipolar derivation localizes focal activity highlighting the phase reversal or cancellation of the signal (zone of isopotentiality).

Legend of the following figure.

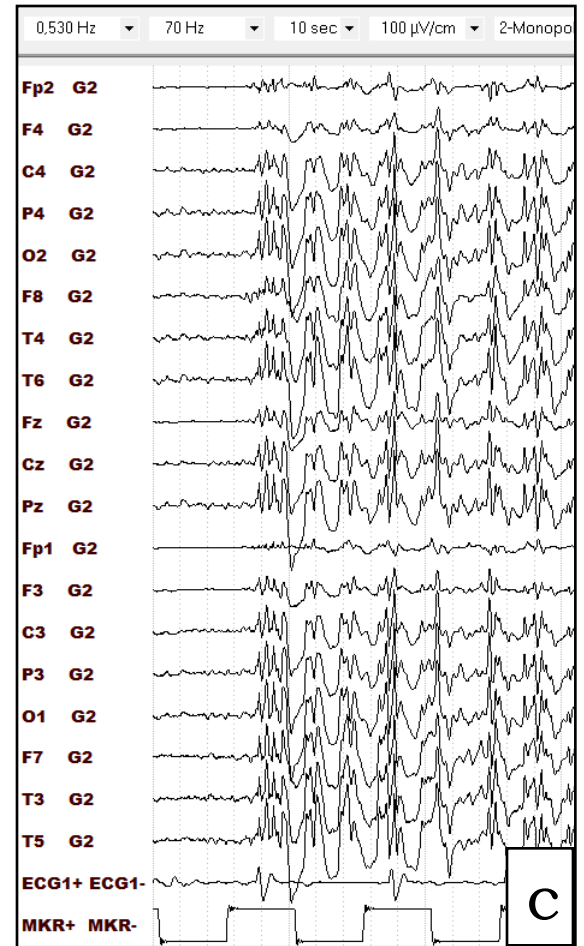
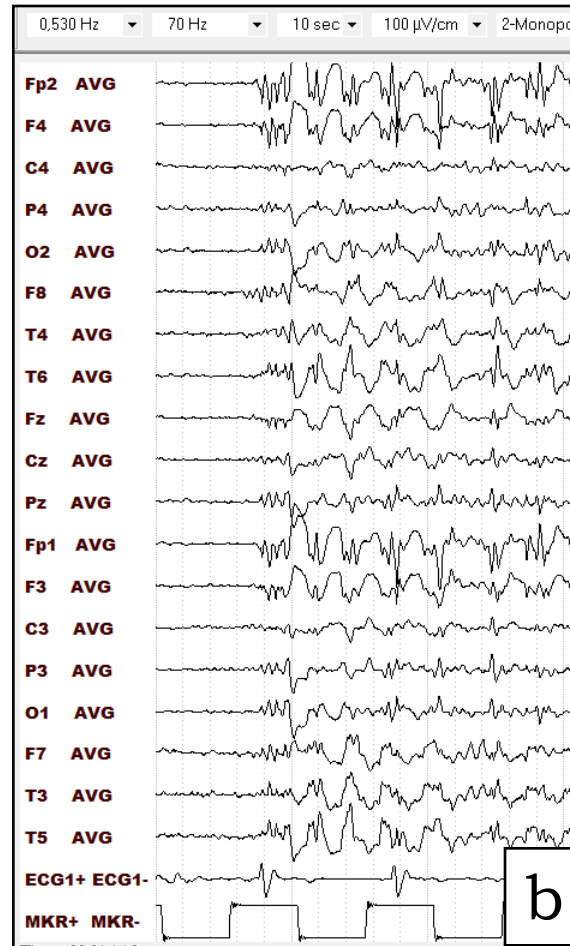
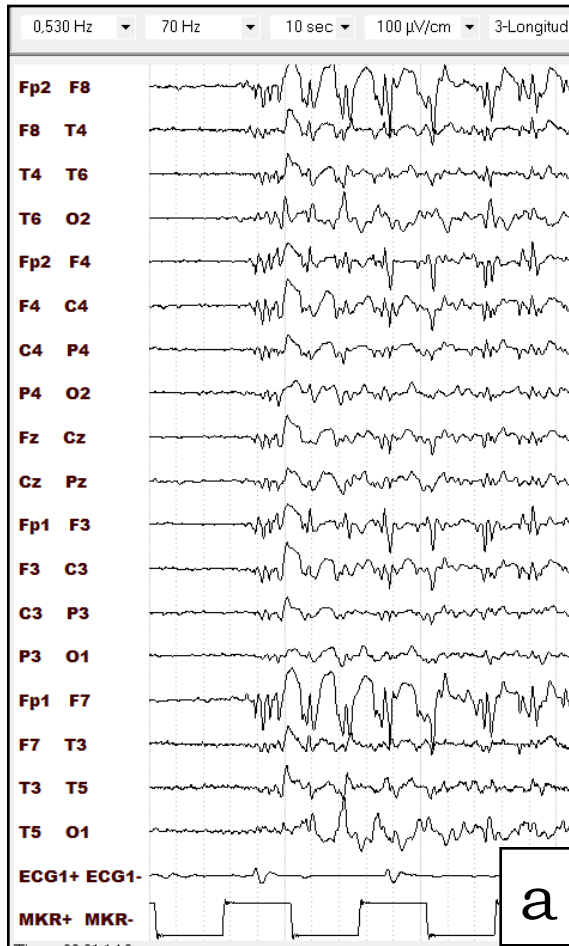
The same epoch of 5 sec of EEG recording shows a right temporal epileptic focus in bipolar, AVG reference, and common active reference derivation. In *a* (bipolar derivation) is evident the phase reversal phenomenon in the first and third channel, with almost total cancellation of spikes in the intervening channel F8-T4 (these electrodes are placed over the focus and their potentials presumably have the same polarity and voltage as input to the differential amplifiers); note the poor spread of spikes to the homologous contralateral areas. The AVG reference derivation (*b*) confirm the higher negative signals at F8 and T4 electrodes; note however that positive signals are present also in Fz, Cz, Fp1, and F3, and negative in F7 and T3. When is used a common active electrode of reference (G2, placed on midline in Fpz) the signal shows the same negative higher voltage in F8 and T4, with the evidence of synchronous lower negative signals contralaterally in F7, T3, T5. However, in this practical example, all three derivations allow to localize the epileptogenic focus with good reliability. (T4, T6, T3, T5 = T8, P8, T7, P7, according to the new nomenclature).



Most of **EEG diffuse activities** are correctly evidenced by bipolar derivations, while average reference (AVG) can be misleading, possibly showing a localized phenomenon as a diffuse one. However, with bipolar derivation, the potential differences between adjacent electrodes are recorded and there is no indication regarding the activity of each electrode with respect to a distant reference point; in addition, bipolar derivation can show a reduction in signal amplitude or, within a widespread pattern, it can underestimate focal amplitude reductions which has the same localization value as spikes. Generally, EEG diffuse activities are better evidenced by a common reference derivation, but the reference electrode should not be affected by the studied activity and so, when this is not possible, it should be placed in a way that allows equal recording of both hemispheres. An extracranial medial reference, for example, can be used.

Legend of the following figure showing a widespread spike-wave discharge, anteriorly predominant, with the three recording methods.

The bipolar (**a**) and AVG reference derivation (**b**) show the real characteristic of discharge, while the common active reference derivation (with G2-reference electrode placed in Fpz) is misleading (**c**). In **c** the contamination reference is very evident and the electrodes with larger signals are precisely those closest to the reference electrode and in them therefore the recorded activity is significantly reduced in voltage. (T4, T6, T3,T5 = T8, P8, T7, P7, according to the new nomenclature).



Montages

A montage is the specific method of electrodes connection to the recording channel of the electroencephalograph. With 21 electrode positions in the 10-20 system and 16 channels on display the number of possible montages is 21. The 10-10 system, with more than 70 electrodes positions, allows the creation of an even greater number of montages, and the modern digital EEG machines allow the display of up to 256 channels.

A wide range of montages, many of which are complex and inadequate, is used in EEG laboratories for routine recordings. This dissimilarity prevents the correct exchange of information between experts in the field. To counteract this, both the International Federation of Clinical Neurophysiology (IFCN) and the American Clinical Neurophysiology Society (ACNS) have published some guidelines which include ad-hoc recommendations.

The montages for routine EEG recording are designated as: Longitudinal Bipolar (LB), Transverse Bipolar (TB) or Referential (R). Montages are designed for 18, 20, 26 and more channels. Here we report a list of the main recommendations drawn from the guidelines:

- no less than 16 channels of simultaneous recording should be used (a larger number of channels would be encouraged);
- at least 21 electrodes should be placed following the 10-20 system (the IFCN recommend at least 25 electrodes, including the inferior temporal chain);
- both bipolar and referential montages should be used for clinical interpretation;
- the electrode derivations of each channel should be clearly identified at the beginning of each montage, so that the pattern of electrode connection is made as simple as possible and easily comprehensible;
- in bipolar derivations, electrode pairs should run in straight lines and their interelectrode distance should be kept equal;
- channel progression must be anterior-posterior.

Guidelines also recommend that a single channel electrocardiogram (ECG) should be included on one EEG channel.

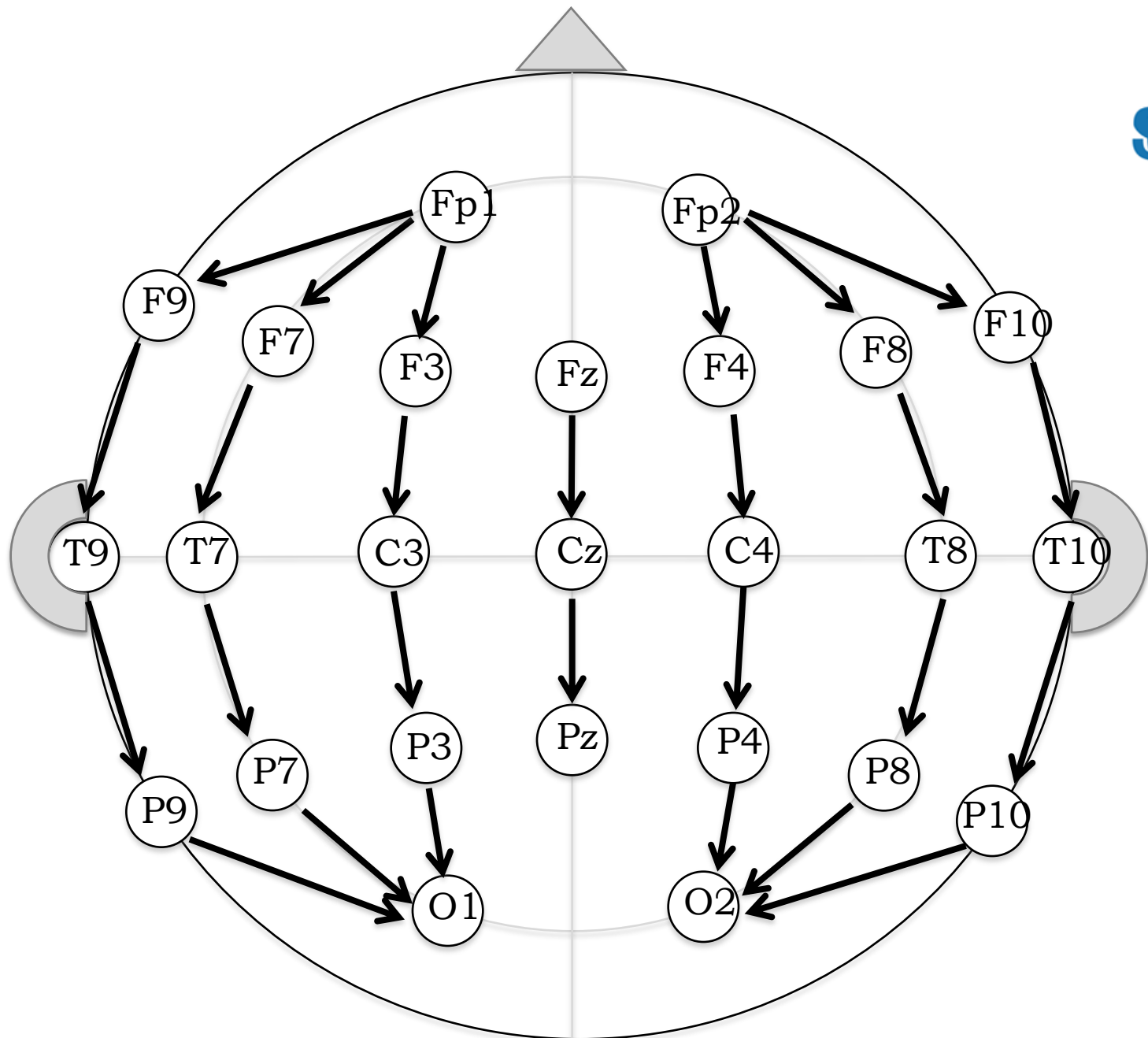
According to IFCN and ACNS, channels obtained by connecting the left-side electrodes should be above the right-sided ones.

This recommendation coincides with the prevailing practice of the vast majority of EEG laboratories in North America and in other areas, but it is not followed in Italy and in many other European countries. In this book, according to the tradition of the Italian and European Neurophysiological academic school and the totality of clinical practice in our country, the right-sided leads are placed above the left-sided leads for either blocks of derivations.

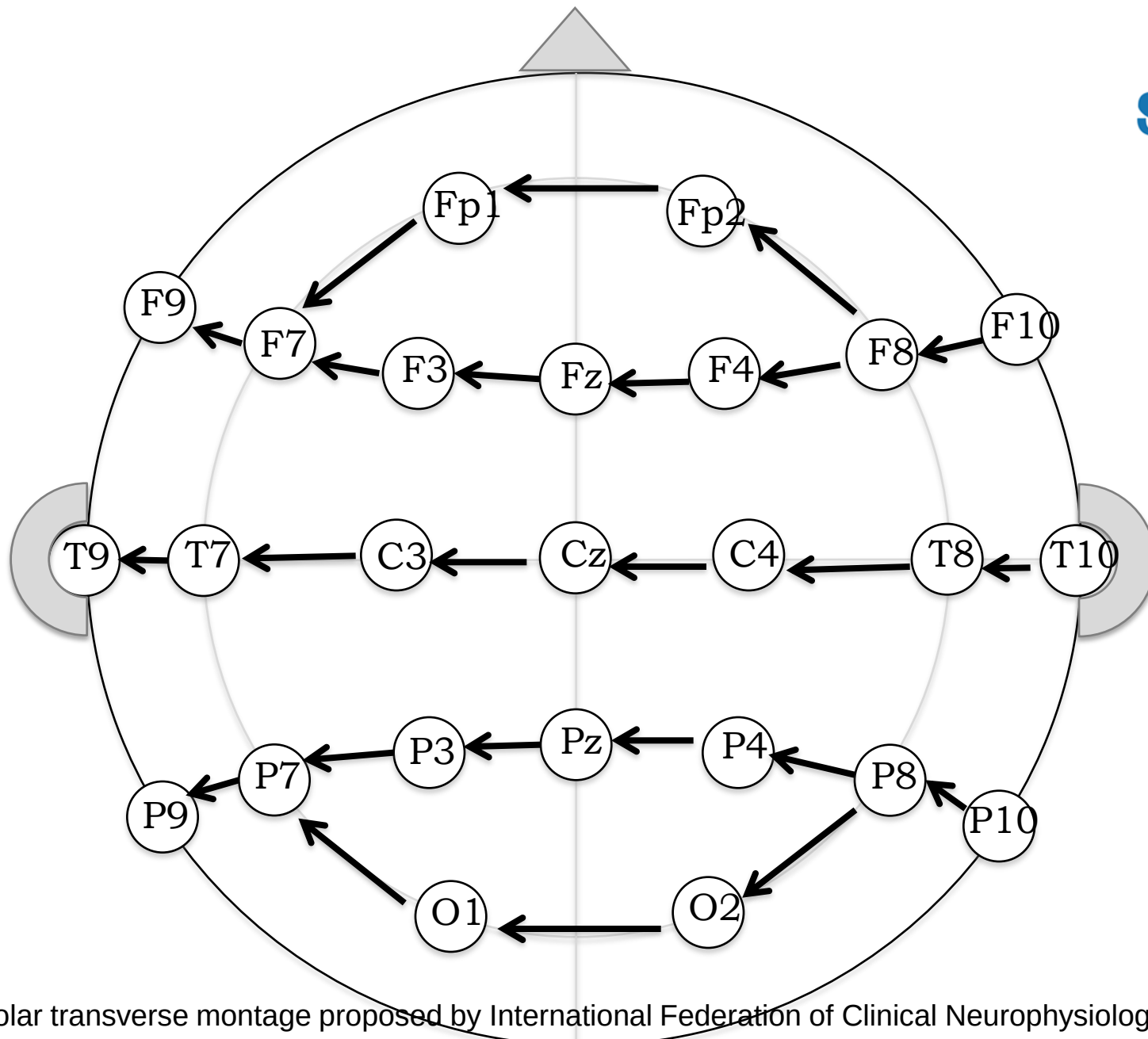
Regarding referential montages the choice of reference is critically important. For the ACNS a midline electrode (as Cz) would be a better choice of reference than A1 or A2. However the referential suggested montages by ACNS establish the right auricular electrode (A2) as reference for the electrodes on the right side, and the left auricular electrode (A1) for the ones on the left side.

Currently in digital electroencephalography, the G2 reference electrode can be used as an “active” reference, placing it on the midline, anteriorly to Fz.

Figures 17 and 18 show suggested bipolar longitudinal and transverse montages with the extended standard array. Table 1 shows bipolar (old and new) and referential suggested montages. According to 10-10 position nomenclature T4, T6, T3, and T5 should be changed in T8, P8, T7, and P7.



New bipolar longitudinal montage proposed by International Federation of Clinical Neurophysiology (IFCN).



New bipolar transverse montage proposed by International Federation of Clinical Neurophysiology (IFCN). Note that the interelectrode distance between the inferior and superior temporal electrodes is shorter (10%) compared to the other inter-electrode distances (20%).

n. ch	LB montage (old)	n. ch	LB montage (new)	n. ch	TB montage (old)	n. ch	TB montage (new)	n. ch	Referential
1	Fp2 - F8	1	Fp2 - F10	1	F8 - Fp2	1	F8 - Fp2	1	Fp2
2	F8 - T8 (F8-T4)	2	F10 - T10	2	Fp2 - Fp1	2	Fp2 - Fp1	2	F10
3	T8 - P8 (T4-T6)	3	T10 - P10	3	Fp1 - F7	3	Fp1 - F7	3	F8
4	P8 - O2(T6-O2)	4	P10 - O2	4	F8 - F4	4	F10 - F8	4	T10
5	Fp2 - F4	5	Fp2 - F8	5	F4 - Fz	5	F8 - F4	5	T8
6	F4 - C4	6	F8 - T8	6	Fz - F3	6	F4 - Fz	6	P10
7	C4 - P4	7	T8 - P8	7	F3 - F7	7	Fz - F3	7	P8
8	P4 - O2	8	P8 - O2	8	T8 - C4 (T4-C4)	8	F3 - F7	8	F4
9	Fz - Cz	9	Fp2 - F4	9	C4 - Cz	9	F7 - F9	9	C4
10	Cz - Pz	10	F4 - C4	10	Cz - C3	10	T10 - T8	10	P4
11	Fp1 - F3	11	C4 - P4	11	C3 - T7 (C3-T3)	11	T8 - C4	11	O2
12	F3 - C3	12	P4 - O2	12	P8 - P4 (T6-P4)	12	C4 - Cz	12	Fp1
13	C3 - P3	13	Fz - Cz	13	P4 - Pz	13	Cz - C3	13	F9
14	P3 - O1	14	Cz - Pz	14	Pz - P3	14	C3 - T7	14	F7
15	Fp1 - F7	15	Fp1 - F3	15	P3 - P7 (P3-T5)	15	T7 - T9	15	T9
16	F7 - T7 (F7-T3)	26	F3 - C3	16	P8 - O2 (T6-O2)	16	P10 - P8	16	T7
17	T7 - P7 (T3-T5)	27	C3 - P3	17	O2 - O1	17	P8 - P4	17	P9
18	P7 - O1 (T5-O1)	18	P3 - O1	18	O1 - P7 (O1-T5)	18	P4 - Pz	18	P7
19	ECG	19	Fp1 - F7	19	ECG	19	Pz - P3	19	F3
		20	F7 - T7			20	P3 - P7	20	C3
		21	T7 - P7			21	P7 - P9	21	P3
		22	P7 - O1			22	P8 - O2	22	O1
		23	Fp1 - F9			23	O2 - O1	23	Fz
		24	F9 - T9			24	O1 - P7	24	Cz
		25	T9 - P9			25	ECG	25	Pz
		26	P9 - O1					26	ECG
		27	ECG						

Bipolar (old and new) and referential suggested montages. According to 10-10 position nomenclature T4, T6, T3, and T5 should be changed in T8, P8, T7, and P7

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